

Comparative Analysis of Churn Prediction Models

A Technical Evaluation Report

Prepared by: Data Science Team

Subject: Telco Customer Churn Strategy

Executive Summary & Business Impact

Data Foundation: This report analyzes a test sample of 1407 customers (374 actual churners).

Primary Insight: The baseline Random Forest classifier optimizes for overall accuracy (78.5%) but fails to identify 52% of actual churners. In contrast, deploying the class-weighted Logistic Regression model increases churn recall to 79.14%, capturing 31% more at-risk customers.

Financial Impact Projection: Assuming an average customer lifetime value of \$780 and a retention cost of \$50, successfully retaining these additionally identified customers yields a projected incremental profit of approximately \$83,950 per 1407 customers evaluated.

This report compares the Baseline Random Forest, a class-weighted Logistic Regression, and an advanced Deep Learning (MLP) architecture using confusion matrices, feature importance, and ROC-AUC analysis.

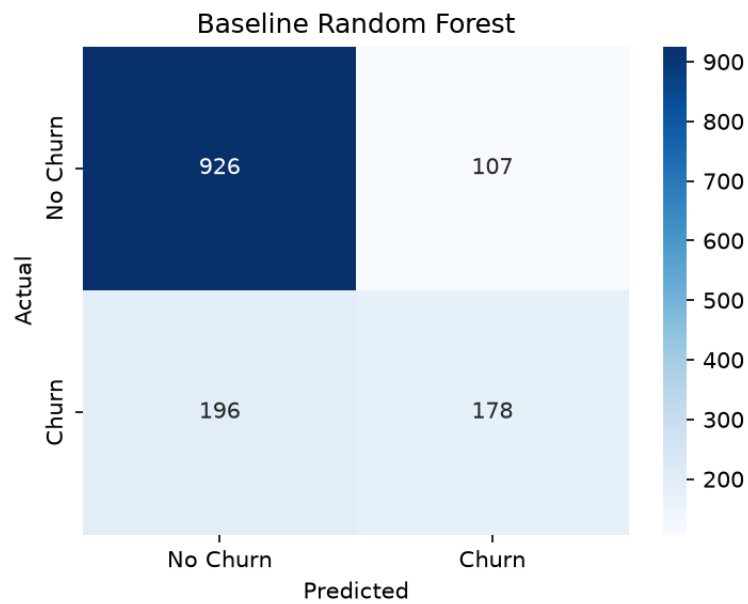
Model 1: Baseline Random Forest

Overall Accuracy: 78.46%

The baseline model achieves high accuracy but suffers from severe class imbalance, predicting 'No Churn' too aggressively.

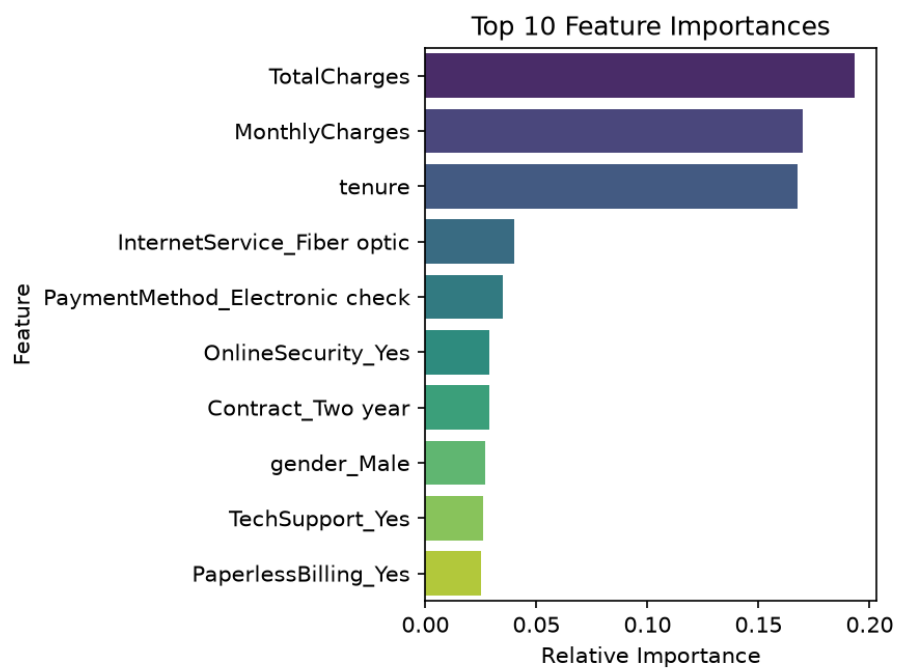
Churn Recall: Only 47.59% of actual churners were identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.83	0.90	0.86	1033.0
Class 1 (Churn)	0.62	0.48	0.54	374.0
Macro Avg	0.72	0.69	0.70	1407.0
Weighted Avg	0.77	0.78	0.77	1407.0



Feature Importance Analysis

The Random Forest model allows us to extract the most critical features driving customer churn. The chart below illustrates the top 10 factors influencing the model's decisions.



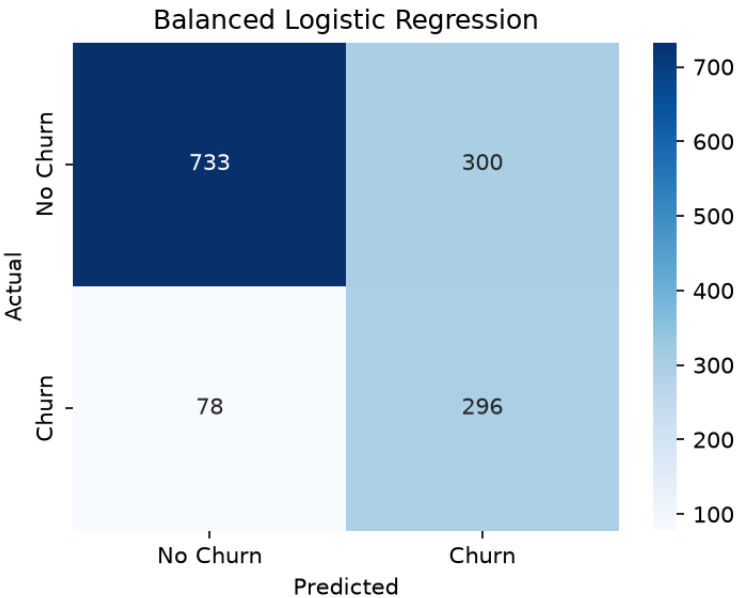
Model 2: Balanced Logistic Regression

Overall Accuracy: 73.13%

By applying 'balanced' class weights, this linear model heavily penalizes the misclassification of the minority class (Churners).

Churn Recall: 79.14% of actual churners were successfully identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.90	0.71	0.80	1033.0
Class 1 (Churn)	0.50	0.79	0.61	374.0
Macro Avg	0.70	0.75	0.70	1407.0
Weighted Avg	0.80	0.73	0.75	1407.0



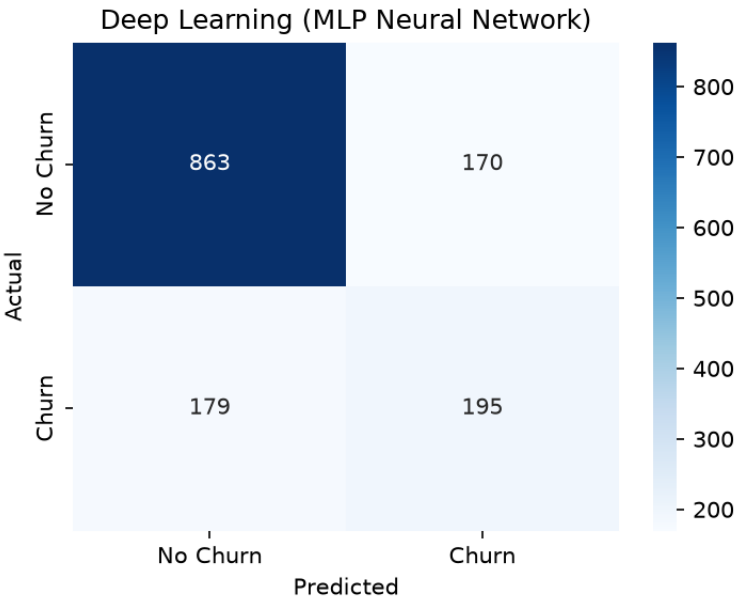
Model 3: Advanced Deep Learning (MLP)

Overall Accuracy: 75.20%

This advanced deep learning architecture utilizes multiple hidden layers to capture highly complex, non-linear feature interactions in the customer data.

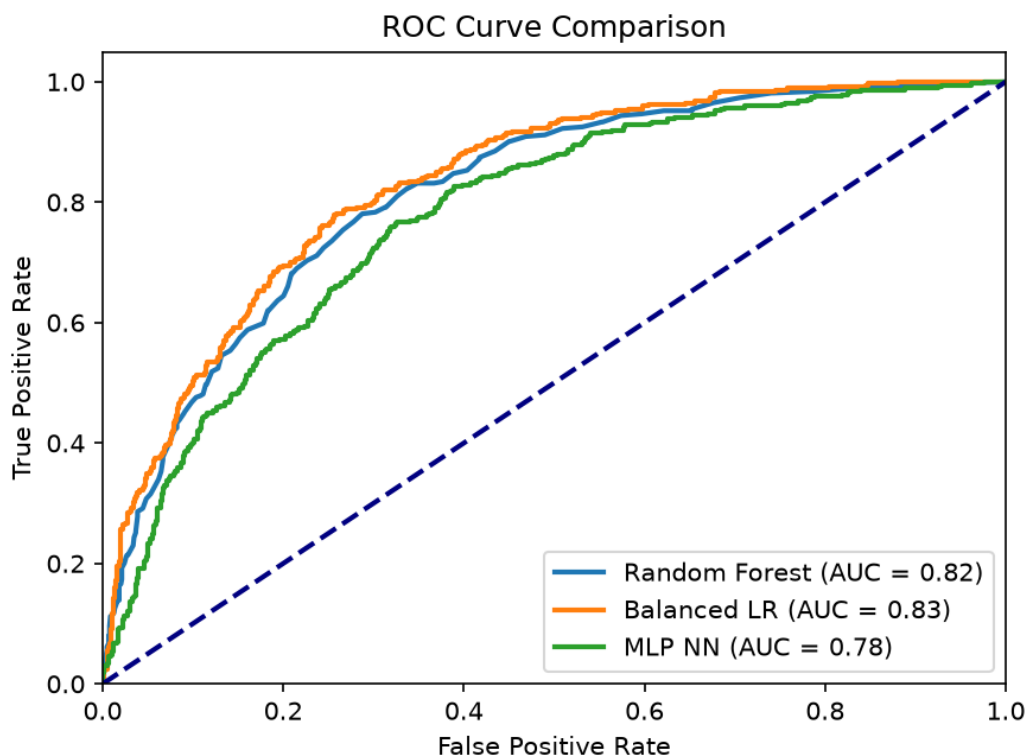
Churn Recall: 52.14% of actual churners were successfully identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.83	0.84	0.83	1033.0
Class 1 (Churn)	0.53	0.52	0.53	374.0
Macro Avg	0.68	0.68	0.68	1407.0
Weighted Avg	0.75	0.75	0.75	1407.0



Model Comparison: ROC-AUC

The Receiver Operating Characteristic (ROC) curve below compares the true positive rate versus the false positive rate across all three models at various threshold settings. The Area Under the Curve (AUC) summarizes the overall performance, where a value closer to 1.0 indicates a superior model.



Strategic Recommendations & Implementation Roadmap

Recommendation 1: Model Deployment (Immediate ROI)

Replace the current baseline Random Forest model with the Balanced Logistic Regression model in production. This will instantly increase churner identification from 47.59% to 79.14%, generating significant ROI through saved accounts.

Recommendation 2: Targeted Retention Campaigns (Phase 1 - 30 days)

Utilize the Feature Importance insights to design segmented retention offers. Customers flagged as high-risk due to 'Month-to-Month' contracts should receive incentives for annual upgrades.

Recommendation 3: Deep Learning Optimization (Phase 2 - 90 days)

While the current Deep Learning (MLP) model achieves a lower recall (52.14%), its ability to model complex interactions remains valuable. Dedicate compute resources to perform an exhaustive hyperparameter search and implement custom loss functions to optimize the MLP specifically for recall.